

UNIT 8: ECOLOGICAL CONCEPTS

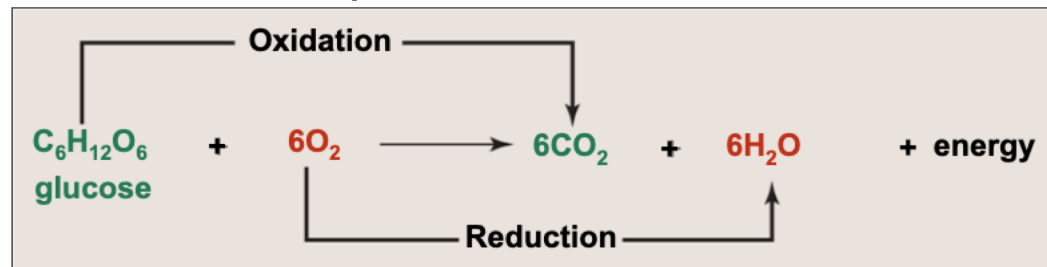
8.5 – CELLULAR RESPIRATION

Cellular Respiration Outline

- Cellular Respiration
 - Outside of the Mitochondria: Glycolysis
 - Outside of the Mitochondria: Fermentation
 - Inside the Mitochondria
 - Metabolic Pool

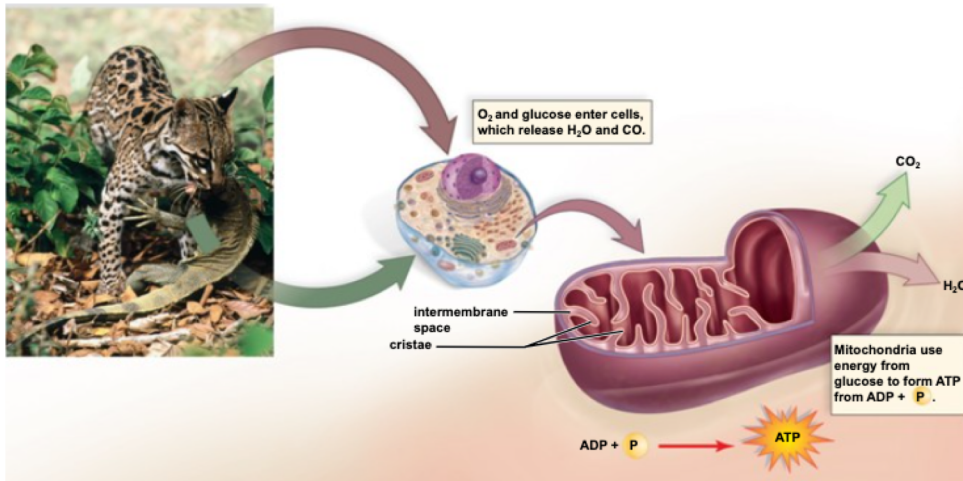
Cellular Respiration

- **Cellular respiration** is a cellular process that breaks down nutrient molecules and the associated production of ATP
- Consumes oxygen and produces carbon dioxide (CO_2)
 - ▣ Cellular respiration is an aerobic process.
- Usually involves the breakdown of glucose to CO_2 and H_2O
 - ▣ Energy is extracted from the glucose molecule:
 - Released step-wise
 - Allows ATP to be produced efficiently
 - ▣ Oxidation-reduction enzymes include NAD^+ and FAD as coenzymes



- ▣ Electrons are removed from substrates and received by oxygen, which combines with H^+ to become water.
- ▣ Glucose is oxidized and O_2 is reduced

Cellular Respiration

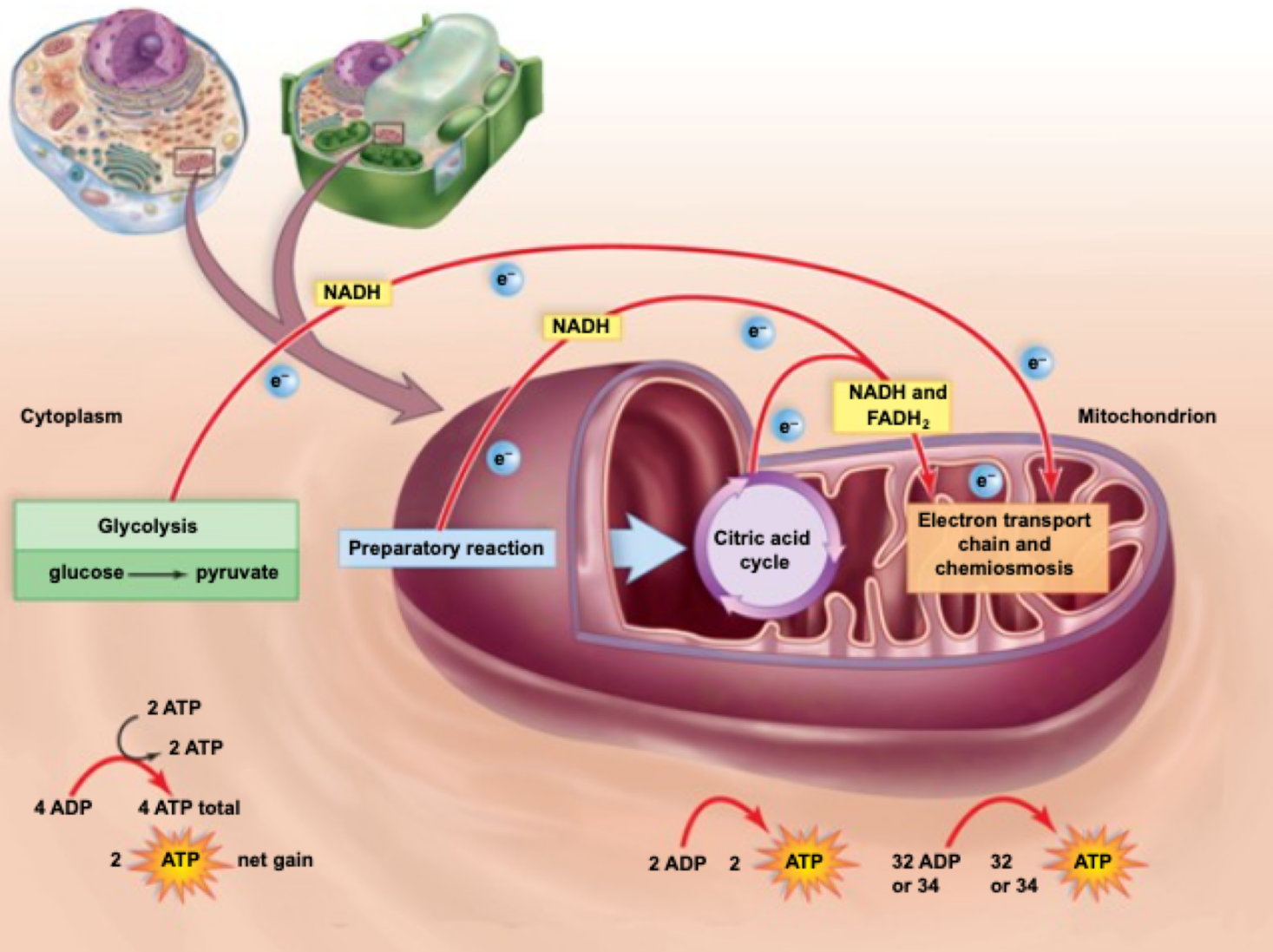


- **NAD⁺** (nicotinamide adenine dinucleotide)
 - ▣ A coenzyme of oxidation-reduction. It can:
 - Oxidize a metabolite by accepting electrons
 - Reduce a metabolite by giving up electrons
 - Each NAD⁺ molecule is used over and over again
- **FAD** (flavin adenine dinucleotide)
 - ▣ Also a coenzyme of oxidation-reduction
 - ▣ Sometimes used instead of NAD⁺
 - ▣ Accepts two electrons and two hydrogen ions (H⁺) to become FADH₂

Cellular Respiration

- Cellular respiration includes four phases:
- **Glycolysis** is the breakdown of glucose into two molecules of pyruvate
 - ▣ Occurs in the cytoplasm
 - ▣ ATP is formed
 - ▣ Does not utilize oxygen (anaerobic)
 - If no oxygen present after pyruvate, undergoes fermentation instead of moving into mitochondria
- **Preparatory (prep) reaction**
 - ▣ Both pyruvates are oxidized and enter the mitochondria
 - ▣ Electron energy is stored in NADH
 - ▣ Two carbons are released as CO_2 (one from each pyruvate)
- **Citric acid cycle**
 - ▣ Occurs in the matrix of the mitochondrion and produces NADH and FADH_2
 - ▣ In series of reactions, it releases 4 carbons as CO_2
 - ▣ Turns twice per glucose molecule (once for each pyruvate)
 - ▣ Produces two immediate ATP molecules per glucose molecule
- **Electron transport chain (ETC)**
 - ▣ Extracts energy from NADH & FADH_2
 - ▣ Passes electrons from higher to lower energy states
 - ▣ Produces 32 or 34 molecules of ATP

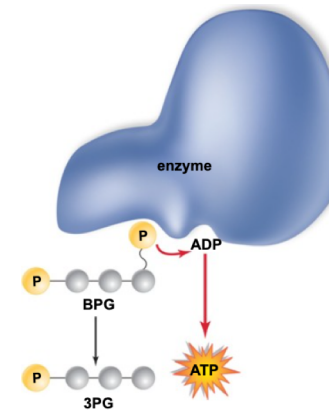
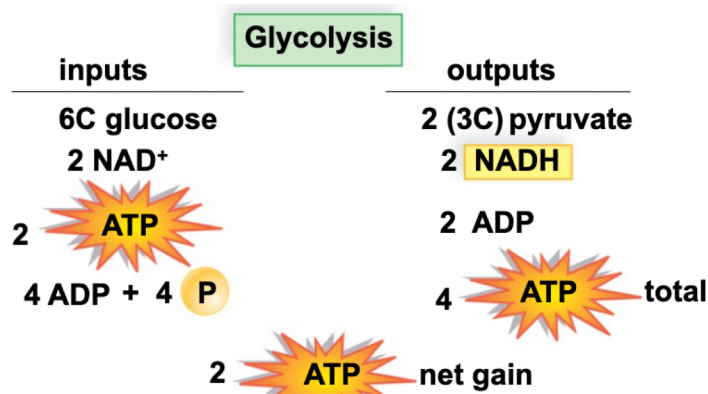
Cellular Respiration



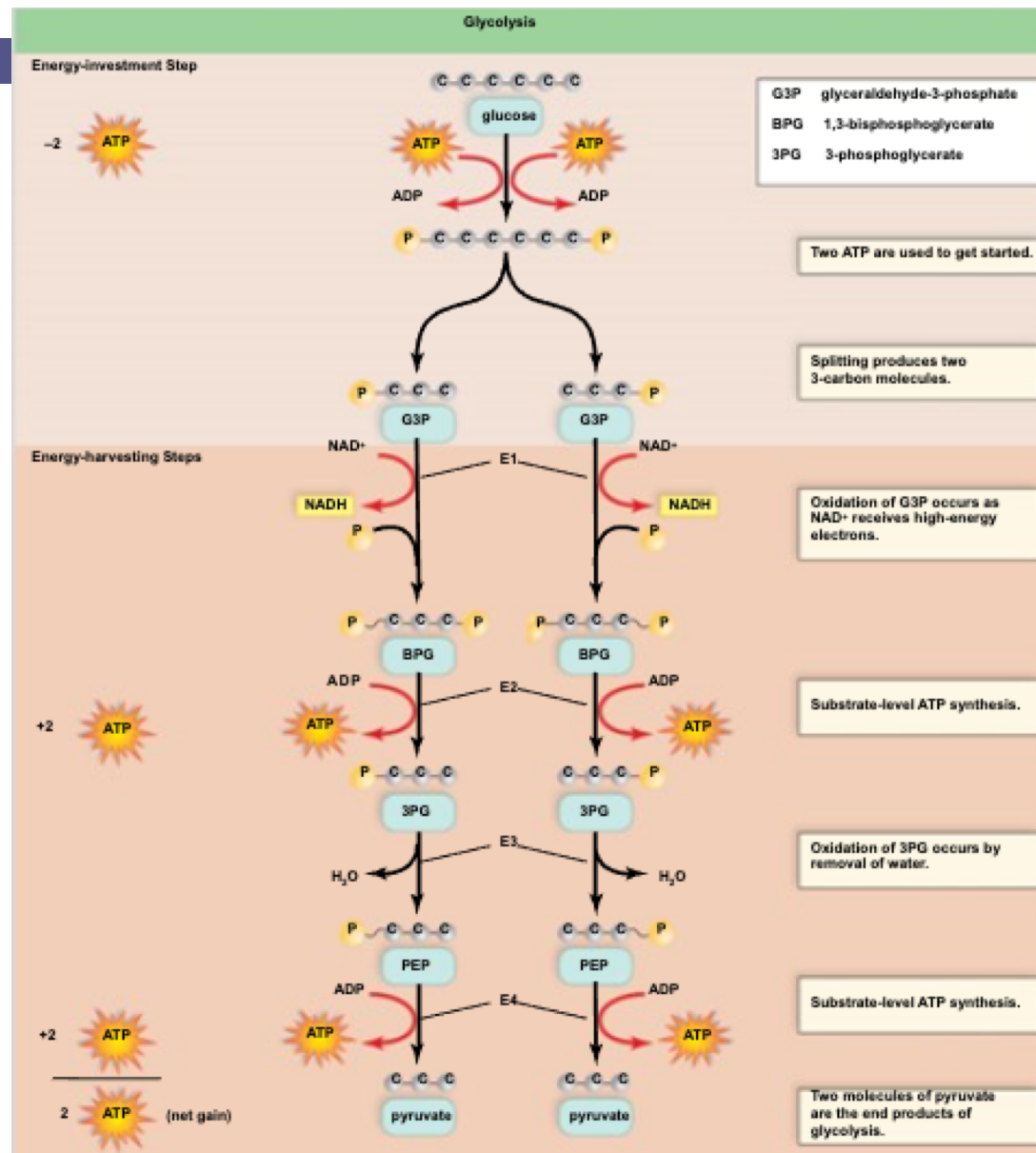
Cellular respiration results in net gain of 36-38 ATP molecules per glucose, depending on the cell

Outside the Mitochondria: Glycolysis

- Glycolysis occurs in cytoplasm outside mitochondria
 - ▣ Energy Investment Step:
 - Two ATP are used to activate glucose
 - Glucose splits into two G3P molecules
 - ▣ Energy Harvesting Step:
 - Oxidation of G3P occurs by removal of electrons and hydrogen ions
 - Two electrons and one hydrogen ion are accepted by NAD^+ resulting in two NADH
 - Four ATP are produced by substrate-level phosphorylation
 - Net gain of two ATP (4 ATP produced - 2 ATP consumed)
 - Both G3Ps converted to pyruvates



Outside the Mitochondria: Glycolysis



Outside the Mitochondria: Fermentation

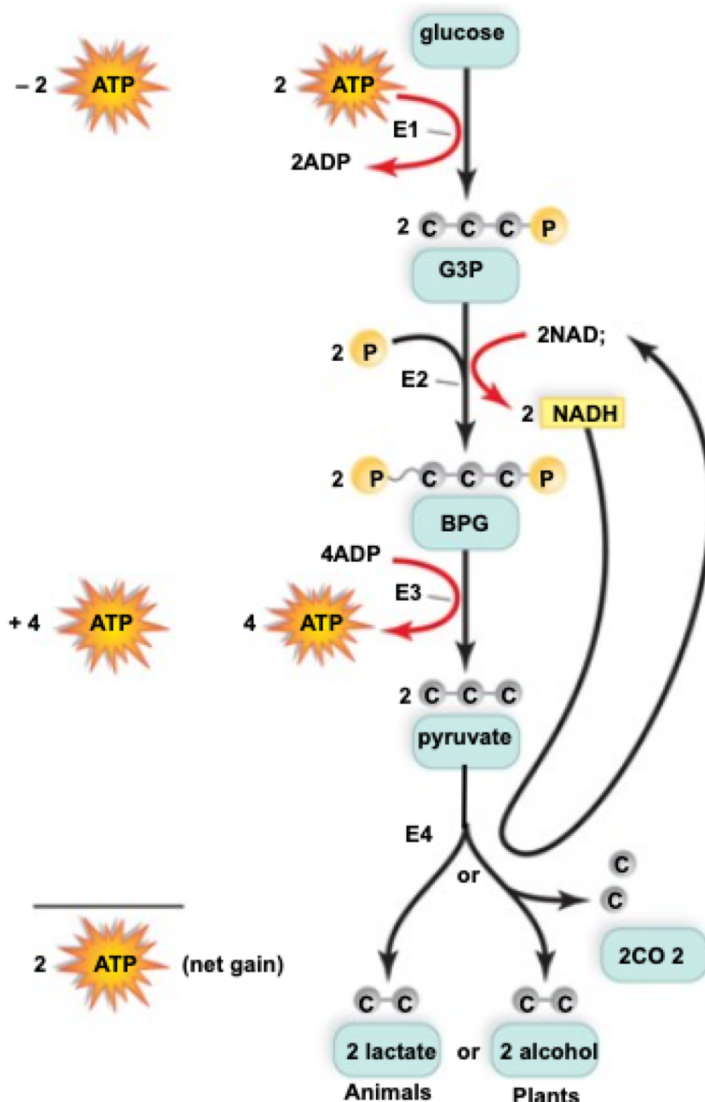
- **Pyruvate** is a pivotal metabolite in cellular respiration
- If O_2 is not available to the cell, **fermentation**, an anaerobic process, occurs in the cytoplasm.
 - ▣ During fermentation, glucose is incompletely metabolized to lactate, or to CO_2 and alcohol (depending on the organism).
- If O_2 is available to the cell, pyruvate enters the mitochondria for aerobic respiration.

Outside the Mitochondria: Fermentation

- Fermentation is an anaerobic process that reduces pyruvate to either lactate or alcohol and CO_2
- NADH transfers its electrons to pyruvate
- Alcoholic fermentation, carried out by yeasts, produces carbon dioxide and ethyl alcohol
 - Used in the production of alcoholic spirits and breads.
- Lactic acid fermentation, carried out by certain bacteria and fungi, produces lactic acid (lactate)
 - Used commercially in the production of cheese, yogurt, and sauerkraut.
- Other bacteria produce chemicals anaerobically, including isopropanol, butyric acid, propionic acid, and acetic acid.



Outside the Mitochondria: Fermentation



Advantages

- Provides a quick burst of ATP energy for muscular activity.

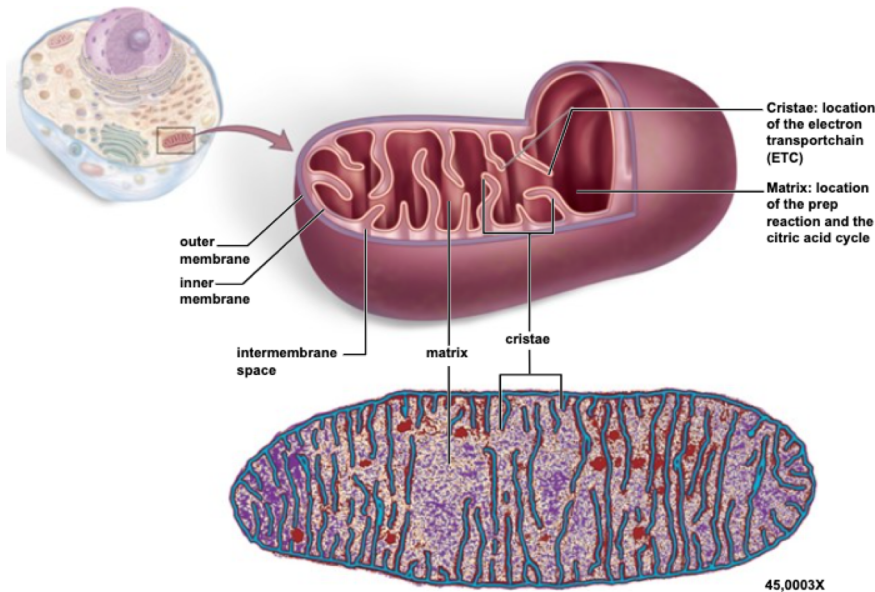
Disadvantages

- Lactate and alcohol are toxic to cells.
- Lactate changes pH and causes muscles to fatigue.
 - Oxygen debt
- Yeast die from the alcohol they produce by fermentation

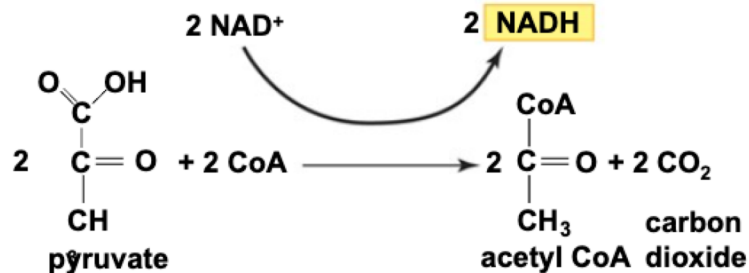
Efficiency of Fermentation

- Two ATP produced per glucose of molecule during fermentation is equivalent to 14.6 kcal.
- Complete oxidation of glucose can yield 686 kcal.
- Only 2 ATP per glucose are produced, compared to 36 or 38 ATP molecules per glucose produced by cellular respiration.

Inside the Mitochondria



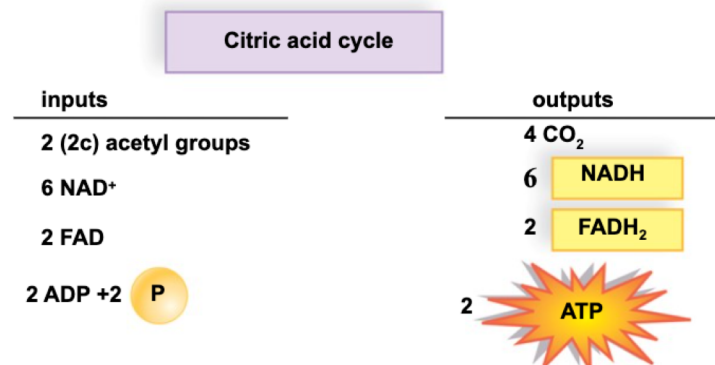
- **The Preparatory (prep) Reaction**
 - ▣ Connects glycolysis to the citric acid cycle
 - ▣ End product of glycolysis, pyruvate, enters the mitochondrial matrix
 - ▣ Pyruvate is converted to a 2-carbon acetyl group
 - Attached to Coenzyme A to form acetyl-CoA
 - Electrons are picked up (as hydrogen atom) by NAD^+
 - CO_2 is released and transported out of mitochondria into the cytoplasm



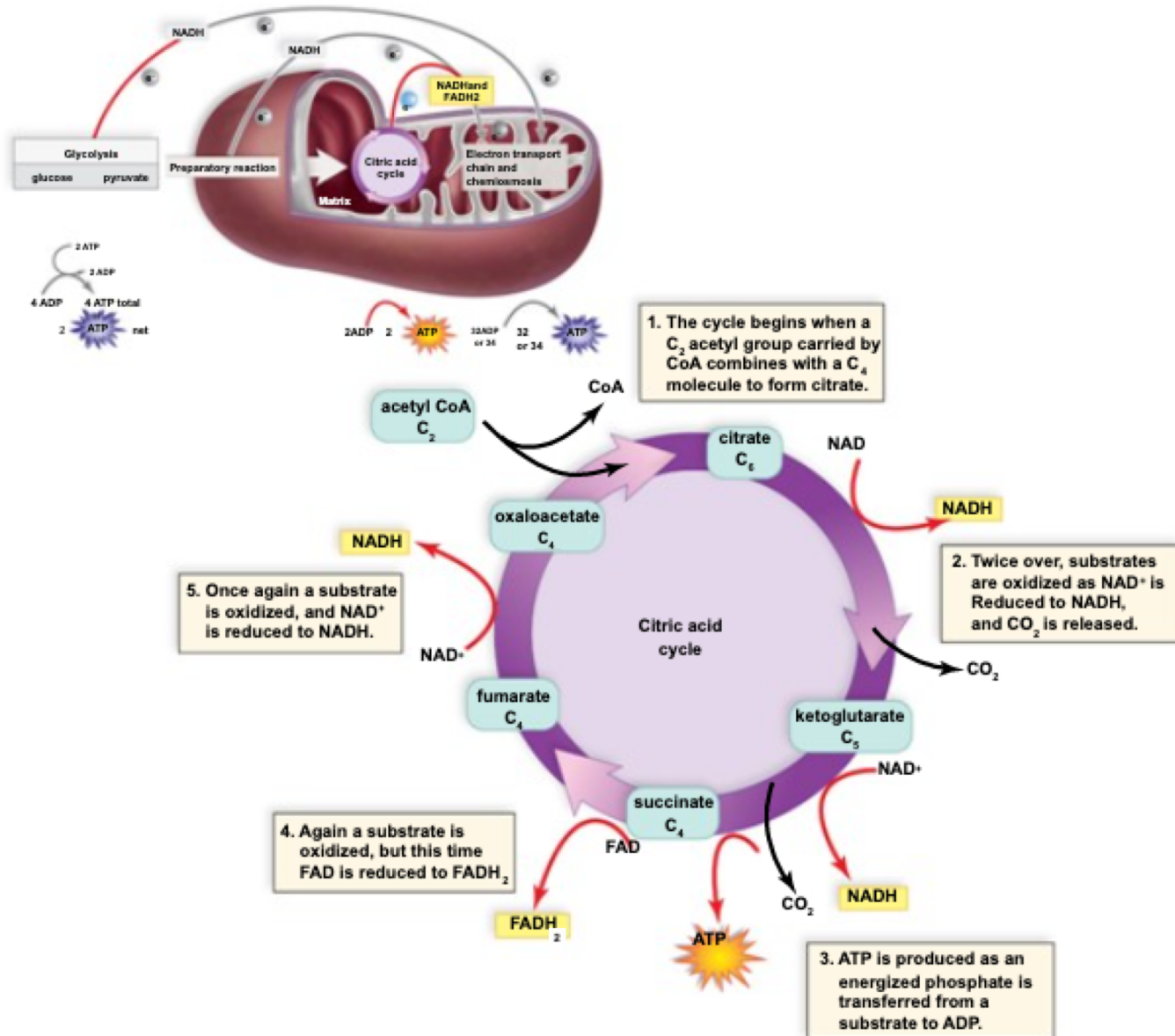
Inside the Mitochondria

□ Citric Acid Cycle

- ▣ Also called the **Krebs Cycle**
- ▣ Occurs in the matrix of mitochondria
- ▣ Begins with the addition of a two-carbon acetyl group (from acetyl-CoA) to a four-carbon molecule (oxaloacetate), forming a six-carbon molecule (citric acid)
- ▣ NAD^+ and FAD capture energy rich electrons
- ▣ ATP is formed by substrate-level phosphorylation
- ▣ Turns twice per glucose molecule (once for each pyruvate)
 - Produces 4 CO_2 , 2 ATP, 6 NADH and 2 FADH_2 per glucose molecule



Inside the Mitochondria



Inside the Mitochondria

□ Electron Transport Chain (ETC)

▣ Location:

- Eukaryotes: cristae of the mitochondria
- Aerobic prokaryotes: plasma membrane

▣ Series of carrier molecules:

- Pass energy-rich electrons successively from one to another
- Complex arrays of protein and cytochrome
 - Cytochromes are proteins with heme groups with central iron atoms

▣ The electron transport chain

- Receives electrons from NADH & FADH₂
- Produces ATP by oxidative phosphorylation

▣ Oxygen serves as the final electron acceptor

- Oxygen combines with hydrogen ions to form water

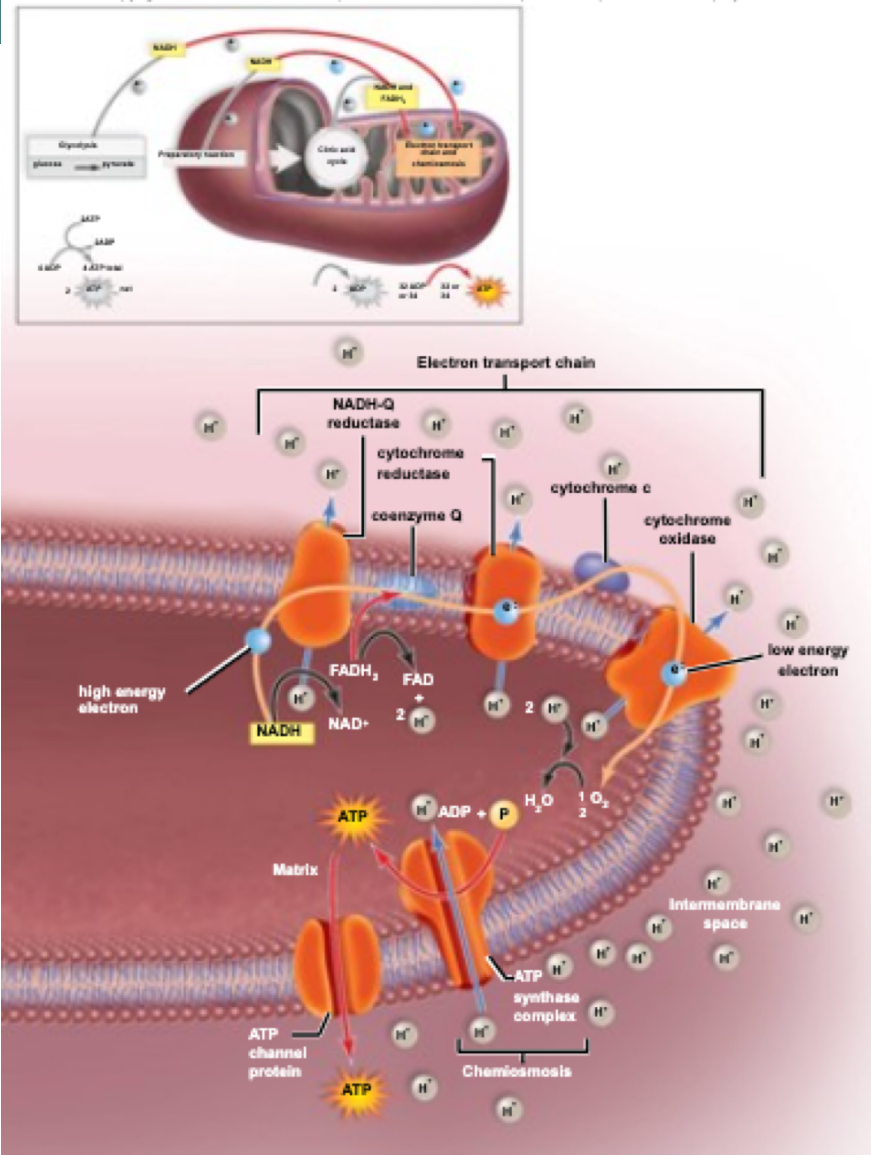
□ The fate of the hydrogens:

- ▣ Hydrogens from **NADH** deliver enough energy to make **3 ATPs**
- ▣ Those from **FADH₂** have only enough for **2 ATPs**
- ▣ “Spent” hydrogens combine with oxygen

□ Recycling of coenzymes increases efficiency

- ▣ Once NADH delivers hydrogens, it returns (as NAD⁺) to pick up more hydrogens
- ▣ However, hydrogens must be combined with oxygen to make water
- ▣ If O₂ is not present, NADH cannot release H⁺
- ▣ No longer recycled back to NAD⁺

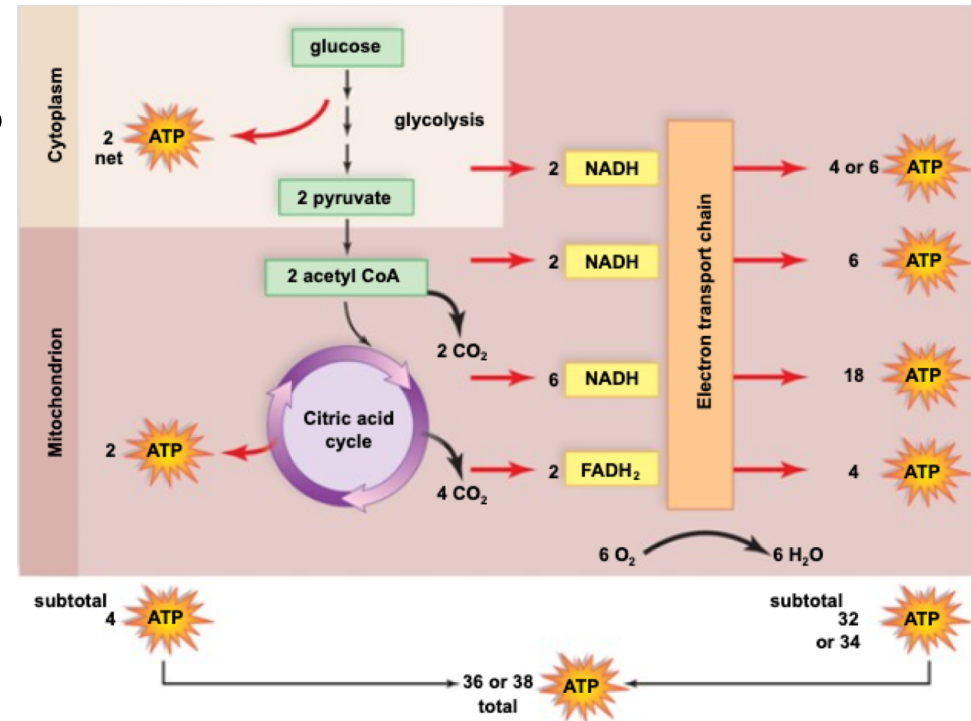
Inside the Mitochondria



- The electron transport chain complexes pump H⁺ from the matrix into the intermembrane space of the mitochondrion
 - ▣ H⁺ therefore becomes more concentrated in the intermembrane space, creating an electrochemical gradient.
- ATP synthase allows H⁺ to flow down its gradient.
- Flow of H⁺ drives the synthesis of ATP from ADP and inorganic phosphate by ATP synthase.
- This process is called **chemiosmosis**
 - ▣ ATP production is linked to the establishment of the H⁺ gradient
- ATP moves out of mitochondria and is used for cellular work
 - ▣ It can be broken down to ADP and inorganic phosphate
 - ▣ These molecules are returned to the mitochondria for more ATP production

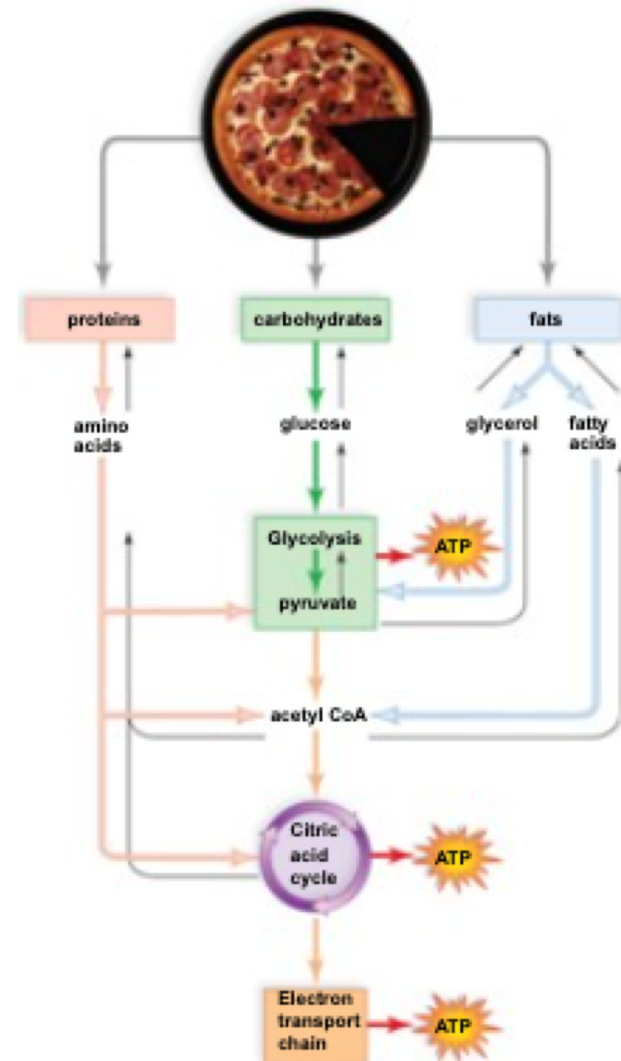
Inside the Mitochondria

- Energy yield from glucose metabolism:
- Net yield per glucose:
 - ▣ From glycolysis – **2 ATP**
 - ▣ From citric acid cycle – **2 ATP**
 - ▣ From electron transport chain – **32 or 34 ATP**
 - # will differ based on how NADH & FADH₂ arrive at ETC
- Energy content:
 - ▣ Reactant (glucose)= 686 kcal
 - ▣ Energy yield (36 ATP)= 263 kcal (7.3kcal/ATP P-bond)
 - ▣ Efficiency is **39%**
 - The rest of the energy from glucose is lost as heat



Metabolic Pool

- Foods:
 - ▣ Sources of energy rich molecules
 - ▣ Carbohydrates, fats, and proteins
- Degradative reactions (**catabolism**) break down molecules
 - ▣ Tend to be exergonic (release energy)
- Synthetic reactions (**anabolism**) build molecules
 - ▣ Tend to be endergonic (consume energy)
- Glucose is broken down in cellular respiration.
- Fat breaks down into glycerol and three fatty acids.
- Amino acids break down into carbon chains and amino groups
 - ▣ Deamination (NH_2 removed) occurs in the liver
 - Results in poisonous ammonia (NH_3)
 - Quickly converted to urea
 - ▣ Different R-groups from amino acids are processed differently
 - ▣ Fragments enter respiratory pathways at many different points



Metabolic Pool

- All metabolic compounds are part of the metabolic pool
- Intermediates from respiratory pathways can be used for anabolism
- Anabolism (synthetic reactions of metabolism):
 - ▣ Carbohydrates
 - Start with acetyl-CoA
 - Basically reverses glycolysis (but different pathway)
 - ▣ Fats
 - G3P converted to glycerol
 - Acetyl groups are connected in pairs to form fatty acids
 - ▣ Proteins:
 - Made up of combinations of 20 different amino acids
 - Some amino acids (11) can be synthesized by adult humans
 - However, other amino acids (9) cannot be synthesized by humans
 - Essential amino acids
 - Must be present in the diet

